Lab 10: Securing the Network with an Intrusion Detection System (IDS)

This lab goes into the processes involved with configuring an Intrusion Detection and Prevention System, scanning for vulnerabilities, and analyzing system scans. We set up our IDS rules and then used the free and open-source software called Snorby as our IDS to scan for vulnerabilities. We then investigated and researched the results of the scan and looked into the high severity events. We found a variety of vulnerabilities including some related to SSH, SNMP, and SQL database servers.

SECTION 1

Part 1: Configure an Intrusion Prevention System (IDS)

```
root@SCO:/etc/nsm/rules# ls
backup                  files.rules         so_rules.rules
black_list.rules        gen-msg.map         stream-events.rules
bpf.conf                http-events.rules   threshold.conf
classification.config   local.rules         tls-events.rules
decoder-events.rules    reference.config    white_list.rules
dns-events.rules        sid-msg.map
downloaded.rules        smtp-events.rules
root@SCO:/etc/nsm/rules#
```

Screenshot 1: Contents of the NSM rules directory

Part 2: Conduct a Vulnerability Scan

Part 3: View the Scan Results

High Severity Events 6 events found

Hotkeys | Classify Event(s) | More Options

Sev.	Sensor	Source IP	Destination IP	Event Signature	Timestamp
1	SCO-eth0:1	172.30.0.10	172.30.0.8	ET TFTP Outbound TFTP Read Request	2:09 AM
1	SCO-eth0:1	172.30.0.10	172.30.0.8	ET TFTP Outbound TFTP Read Request	2:09 AM
1	SCO-eth0:1	172.30.0.10	172.30.0.8	ET TFTP Outbound TFTP Read Request	2:09 AM
1	SCO-eth0:1	172.30.0.10	172.30.0.8	ET TFTP Outbound TFTP Read Request	2:09 AM
1	SCO-eth0:1	172.30.0.10	172.30.0.8	ET TFTP Outbound TFTP Read Request	2:09 AM
1	SCO-eth0:1	172.30.0.10	172.30.0.8	ET TFTP Outbound TFTP Read Request	2:09 AM

Screenshot 2: High-severity events detected by Snorby

Top 15 Signatures

Signature Name	Percentage	Event Count
ET SCAN Potential SSH Scan OUTBOUND	15.22%	7
ET TFTP Outbound TFTP Read Request	13.04%	6
GPL SNMP public access udp	10.87%	5
ET SCAN Potential SSH Scan	8.7%	4
GPL RPC xdmcp info query	6.52%	3
GPL MISC UPnP malformed advertisement	6.52%	3
ET POLICY Suspicious inbound to PostgreSQL port 5432	4.35%	2
ET POLICY Suspicious inbound to mySQL port 3306	4.35%	2
ET POLICY Suspicious inbound to MSSQL port 1433	4.35%	2
ET POLICY MS Remote Desktop Administrator Login Request	4.35%	2
ET POLICY Suspicious inbound to mSQL port 4333	4.35%	2
ET POLICY Suspicious inbound to Oracle SQL port 1521	4.35%	2
ET POLICY RDP connection confirm	4.35%	2
ET SCAN Potential VNC Scan 5900-5920	2.17%	1
ET SCAN Potential VNC Scan 5800-5820	2.17%	1
ET INFO Session Traversal Utilities for NAT (STUN Binding Requ...	2.17%	1
ET DOS Possible SSDP Amplification Scan in Progress	2.17%	1

Screenshot 3: Top 15 signatures in the pdf report from Snorby

SECTION 2

Part 1: Configure an Intrusion Prevention System (IDS)

```
root@SCO:~# cat local.rules
alert tcp any any -> $HOME_NET 22 (msg:"gianna SSH connection attempt"; sid:1000
002; rev:1;)
```

Screenshot 4: Contents of the local.rules file

Signature: ET SCAN Potential SSH Scan

```
alert tcp $EXTERNAL_NET any -> $HOME_NET 22 (msg:"ET SCAN Potential SSH Scan";
flags:S,12; threshold: type both, track by_src, count 5, seconds 120;
reference:url,en.wikipedia.org/wiki/Brute_force_attack;
reference:url,doc.emergingthreats.net/2001219; classtype:attempted-recon;
sid:2001219; rev:19;)
```

Screenshot 5: Rule information for the event created by the SSH connection

Part 2: Conduct a Vulnerability Scan

Part 3: View the Scan Results

Top 15 Signatures

Signature Name	Percentage	Event Count
ET SCAN Potential SSH Scan OUTBOUND	15.22%	14
ET TFTP Outbound TFTP Read Request	13.04%	12
GPL SNMP public access udp	10.87%	10
ET SCAN Potential SSH Scan	8.7%	8
GPL RPC xdmcp info query	6.52%	6
GPL MISC UPnP malformed advertisement	6.52%	6
ET POLICY Suspicious inbound to PostgreSQL port 5432	4.35%	4
ET POLICY Suspicious inbound to mySQL port 3306	4.35%	4
ET POLICY Suspicious inbound to MSSQL port 1433	4.35%	4
ET POLICY MS Remote Desktop Administrator Login Request	4.35%	4
ET POLICY Suspicious inbound to mSQL port 4333	4.35%	4
ET POLICY Suspicious inbound to Oracle SQL port 1521	4.35%	4
ET POLICY RDP connection confirm	4.35%	4
ET SCAN Potential VNC Scan 5900-5920	2.17%	2
ET SCAN Potential VNC Scan 5800-5820	2.17%	2
ET INFO Session Traversal Utilities for NAT (STUN Binding Requ...	2.17%	2
ET DOS Possible SSDP Amplification Scan in Progress	2.17%	2

Screenshot 6: Top 15 signatures in the pdf report from Snorby

Research and describe three of the signatures alerts captured in the Snorby report:

ET SCAN Potential SSH Scan OUTBOUND

This alert warns administrators about potential internal hosts scanning external networks for SSH services, which could be a sign of malicious intent and should be investigated further to prevent security breaches. This behavior could be indicative of reconnaissance or attempts to exploit vulnerable SSH servers for unauthorized access or other malicious purposes.

GPL SNMP public access udp

This alert indicates potential unauthorized access or misuse of SNMP via UDP, meaning that the SNMP service on a network device is configured to allow access without proper authentication or with default or weak community strings. This configuration can expose the device to potential security risks, as attackers can use SNMP to gather sensitive information or even modify device settings if not properly secured. This could lead to unauthorized access or exploitation of network devices if not addressed.

ET POLICY Suspicious Inbound to Oracle SQL port 1521

This alert indicates potentially suspicious network traffic directed toward port 1521, which is commonly associated with Oracle SQL database servers. There may be network traffic attempting to establish connections with Oracle SQL databases on port 1521 in a suspicious manner. The alert may indicate unauthorized access attempts, scanning activity, reconnaissance attempts, or other malicious activity like data theft, manipulation, or sabotage.